

0.1 Mathematical Foundations

0.1.1 Rigid-Body Dynamics

$$\mathbf{S}(\mathbf{k}) = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix} \quad (1)$$

$$\mathbf{L}(\mathbf{k}) = \begin{bmatrix} k_1 & k_2 & k_3 & 0 & 0 & 0 \\ 0 & k_1 & 0 & k_2 & k_3 & 0 \\ 0 & 0 & k_1 & 0 & k_2 & k_3 \end{bmatrix} \quad (2)$$

$$\mathbf{A}^k = \begin{bmatrix} \mathbf{a}^k & \mathbf{S}(\dot{\boldsymbol{\omega}}^k) + \mathbf{S}(\boldsymbol{\omega}^k)\mathbf{S}(\boldsymbol{\omega}^k) & \mathbf{0}_{3 \times 6} \\ \mathbf{0}_{3 \times 1} & -\mathbf{S}(\mathbf{a}^k) & \mathbf{L}(\dot{\boldsymbol{\omega}}^k) + \mathbf{S}(\boldsymbol{\omega}^k)\mathbf{L}(\boldsymbol{\omega}^k) \end{bmatrix} \in \mathbb{R}^{6 \times 10} \quad (3)$$

0.1.2 Log-Cholesky

$$\boldsymbol{\phi} = [m, h_x, h_y, h_z, I_{xx}, I_{xy}, I_{xz}, I_{yy}, I_{yz}, I_{zz}]^\top \quad (4)$$

$$\boldsymbol{\theta} = [\alpha, d_1, d_2, d_3, s_{12}, s_{23}, s_{13}, t_1, t_2, t_3]^\top \in \mathbb{R}^{10} \quad (5)$$

$\boldsymbol{\phi}$ to $\boldsymbol{\theta}$

$$\boldsymbol{\Sigma} = \frac{1}{2} \text{tr}(\bar{\mathbf{I}}) \mathbf{1}_3 - \bar{\mathbf{I}} \quad (6)$$

$$\mathbf{J} = \begin{bmatrix} \boldsymbol{\Sigma} & \mathbf{h} \\ \mathbf{h}^\top & m \end{bmatrix} \quad (7)$$

$$\mathbf{U} = \text{chol}(\mathbf{J}) = e^\alpha \begin{bmatrix} e^{d_1} & s_{12} & s_{13} & t_1 \\ 0 & e^{d_2} & s_{23} & t_2 \\ 0 & 0 & e^{d_3} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (8)$$

$$\alpha = \log \mathbf{U}_{[4,4]}, \quad (9)$$

$$d_1 = \log (\mathbf{U}_{[1,1]}/e^\alpha), \quad (10)$$

$$d_2 = \log (\mathbf{U}_{[2,2]}/e^\alpha), \quad (11)$$

$$d_3 = \log (\mathbf{U}_{[3,3]}/e^\alpha), \quad (12)$$

$$s_{12} = \mathbf{U}_{[1,2]}/e^\alpha, \quad (13)$$

$$s_{23} = \mathbf{U}_{[2,3]}/e^\alpha, \quad (14)$$

$$s_{13} = \mathbf{U}_{[1,3]}/e^\alpha, \quad (15)$$

$$t_1 = \mathbf{U}_{[1,4]}/e^\alpha, \quad (16)$$

$$t_2 = \mathbf{U}_{[2,4]}/e^\alpha, \quad (17)$$

$$t_3 = \mathbf{U}_{[3,4]}/e^\alpha \quad (18)$$

θ to ϕ

$$m = e^{2\alpha} (t_1^2 + t_2^2 + t_3^2 + 1), \quad (19)$$

$$h_x = e^{2\alpha} t_1 e^{d_1}, \quad (20)$$

$$h_y = e^{2\alpha} (t_1 s_{12} + t_2 e^{d_2}), \quad (21)$$

$$h_z = e^{2\alpha} (t_1 s_{13} + t_2 s_{23} + t_3 e^{d_3}), \quad (22)$$

$$I_{xx} = e^{2\alpha} (s_{12}^2 + e^{2d_2} + s_{13}^2 + s_{23}^2 + e^{2d_3}), \quad (23)$$

$$I_{yy} = e^{2\alpha} (e^{2d_1} + s_{13}^2 + s_{23}^2 + e^{2d_3}), \quad (24)$$

$$I_{zz} = e^{2\alpha} (e^{2d_1} + s_{12}^2 + e^{2d_2}), \quad (25)$$

$$I_{xy} = -e^{2\alpha} s_{12} e^{d_1}, \quad (26)$$

$$I_{xz} = -e^{2\alpha} s_{13} e^{d_1}, \quad (27)$$

$$I_{yz} = -e^{2\alpha} (s_{12} s_{13} + s_{23} e^{d_2}) \quad (28)$$

$$(\hat{\theta}, \hat{\mathbf{b}}) = \arg \min_{\theta, \mathbf{b}} \sum_{k=1}^N \left\| \mathbf{G} (\mathbf{w}^k - \mathbf{A}^k \phi(\theta) - \mathbf{b}) \right\|^2 + \lambda^2 \left\| \theta - \hat{\theta}_{\text{CAD}} \right\|^2 \quad (29)$$

0.1.3 Analytical Jacobian

$$\mathbf{r}(\theta, \mathbf{b}) = \begin{bmatrix} \mathbf{G} (\mathbf{w}^k - \mathbf{A}^k \phi(\theta) - \mathbf{b}) \\ \lambda (\theta - \theta_{\text{CAD}}) \end{bmatrix} \quad (30)$$

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \mathbf{r}}{\partial \theta} & \frac{\partial \mathbf{r}}{\partial \mathbf{b}} \end{bmatrix} = \begin{bmatrix} -\mathbf{G} \mathbf{A}^k \frac{\partial \phi}{\partial \theta} & -\mathbf{G} \\ \lambda \mathbf{I} & \mathbf{0} \end{bmatrix} \quad (31)$$

$$\frac{\partial \phi}{\partial \theta} = \begin{bmatrix} 2m & 0 & 0 & 0 & 0 & 0 & 0 & 2e^{2\alpha} t_1 & 2e^{2\alpha} t_2 & 2e^{2\alpha} t_3 \\ 2h_x & e^{2\alpha} t_1 e^{d_1} & 0 & 0 & 0 & 0 & 0 & e^{2\alpha} e^{d_1} & 0 & 0 \\ 2h_y & 0 & e^{2\alpha} t_2 e^{d_2} & 0 & e^{2\alpha} t_1 & 0 & 0 & e^{2\alpha} s_{12} & e^{2\alpha} e^{d_2} & 0 \\ 2h_z & 0 & 0 & e^{2\alpha} t_3 e^{d_3} & 0 & e^{2\alpha} t_2 & e^{2\alpha} t_1 & e^{2\alpha} s_{13} & e^{2\alpha} s_{23} & e^{2\alpha} e^{d_3} \\ 2I_{xx} & 0 & 2e^{2\alpha} e^{2d_2} & 2e^{2\alpha} e^{2d_3} & 2e^{2\alpha} s_{12} & 2e^{2\alpha} s_{23} & 2e^{2\alpha} s_{13} & 0 & 0 & 0 \\ 2I_{xy} & -e^{2\alpha} s_{12} e^{d_1} & 0 & 0 & -e^{2\alpha} e^{d_1} & 0 & 0 & 0 & 0 & 0 \\ 2I_{xz} & -e^{2\alpha} s_{13} e^{d_1} & 0 & 0 & 0 & 0 & -e^{2\alpha} e^{d_1} & 0 & 0 & 0 \\ 2I_{yy} & 2e^{2\alpha} e^{2d_1} & 0 & 2e^{2\alpha} e^{2d_3} & 0 & 0 & 2e^{2\alpha} s_{13} & 0 & 0 & 0 \\ 2I_{yz} & 0 & -e^{2\alpha} s_{23} e^{d_2} & 0 & -e^{2\alpha} s_{13} & -e^{2\alpha} e^{d_2} & -e^{2\alpha} s_{12} & 0 & 0 & 0 \\ 2I_{zz} & 2e^{2\alpha} e^{2d_1} & 2e^{2\alpha} e^{2d_2} & 0 & 2e^{2\alpha} s_{12} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (32)$$

0.1.4 Condition Number

$$\kappa = \frac{s_1}{s_6} \quad (33)$$

$$\kappa_{\text{eff}} = \sqrt{\frac{s_1^2 + \lambda^2}{s_{10}^2 + \lambda^2}} \quad (34)$$

Table 1: Load cell calibrated ranges ($\pm$)

F_x	F_y	F_z	T_x	T_y	T_z
950 N	950 N	1900 N	40 N·m	40 N·m	40 N·m

Table 2: Measurement uncertainty (95% confidence, percent of full-scale load)

F_x	F_y	F_z	T_x	T_y	T_z
1.50%	1.50%	1.00%	1.25%	1.25%	1.25%

0.1.5 Residual Weights

The noise standard deviations are computed as range $\times$ uncertainty:

$$\sigma_{F_x} = \sigma_{F_y} = 950 \times 0.015, \quad (35)$$

$$\sigma_{F_z} = 1900 \times 0.010, \quad (36)$$

$$\sigma_{T_x} = \sigma_{T_y} = \sigma_{T_z} = 40 \times 0.0125 \quad (37)$$

and the residual weighting is normalised by the torque noise:

$$w_i = \frac{\sigma_{T_x}}{\sigma_i} \quad (38)$$

0.1.6 Savitzky-Golay

$$\dot{x}^{(p)} = \frac{p!}{(-\Delta t)^p} (x * g_{p+1}) \quad (39)$$

0.1.7 IMU Data in LC Frame

$$\mathbf{a} = {}^{\text{LC}}\mathbf{R}_{\text{IMU}} \mathbf{a}_{\text{IMU}}, \quad (40)$$

$$\boldsymbol{\omega} = {}^{\text{LC}}\mathbf{R}_{\text{IMU}} \boldsymbol{\omega}_{\text{IMU}}, \quad (41)$$

$$\dot{\boldsymbol{\omega}} = {}^{\text{LC}}\mathbf{R}_{\text{IMU}} \dot{\boldsymbol{\omega}}_{\text{IMU}} \quad (42)$$

$$\mathbf{a}_{\text{LC}} = \mathbf{a} + \dot{\boldsymbol{\omega}} \times (-{}^{\text{LC}}\mathbf{r}_{\text{LC,IMU}}) + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times (-{}^{\text{LC}}\mathbf{r}_{\text{LC,IMU}})) \quad (43)$$

0.1.8 Compute Point Cloud Alignment

Uses Kabsch Algorithm. Subtract the mean from measured and reference point clouds

$$\tilde{P}_{\text{ref},i} = P_{\text{ref},i} - \frac{1}{N} \sum_{i=1}^N P_{\text{ref},i}, \quad (44)$$

$$\tilde{P}_{\text{meas},i} = P_{\text{meas},i} - \frac{1}{N} \sum_{i=1}^N P_{\text{meas},i} \quad (45)$$

Compute cross-covariance matrix is formed and decomposed

$$\mathbf{H} = \tilde{\mathbf{P}}_{\text{ref}} \tilde{\mathbf{P}}_{\text{meas}}^\top = \mathbf{U} \mathbf{S} \mathbf{V}^\top \in \mathbb{R}^{3 \times 3} \quad (46)$$

The optimal rotation is then

$$\mathbf{R} = \mathbf{V} \mathbf{U}^\top \quad (47)$$

Internet says swap sign of third column if determinant is negative...

$$\mathbf{V}[:,3] \leftarrow \begin{cases} -\mathbf{V}[:,3] & \det(\mathbf{R}) < 0 \\ \mathbf{V}[:,3] & \text{otherwise} \end{cases} \quad (48)$$

compute translation vector

$$\mathbf{t} = \bar{\mathbf{p}}_{\text{meas}} - \mathbf{R} \bar{\mathbf{p}}_{\text{ref}} \quad (49)$$

full transform

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \quad (50)$$

0.2 Handles

0.2.1 CAD Prior

I_{axial} is inertia about solid rod axis. I_{radial} is inertia about central diameter.

$$I_{\text{axial}} = \frac{1}{2} m r_h^2, \quad (51)$$

$$I_{\text{radial}} = \frac{1}{12} m (3r_h^2 + L^2) \quad (52)$$

The actual handle is approximately long in $\mathbf{x}$ but rotated by -45 deg.

$$\mathbf{I}_{\text{com}} = \mathbf{R}_z(\theta) \begin{bmatrix} I_{\text{axial}} & 0 & 0 \\ 0 & I_{\text{radial}} & 0 \\ 0 & 0 & I_{\text{radial}} \end{bmatrix} \mathbf{R}_z(\theta)^\top \quad (53)$$

Distance z from sensor frame to center of rod

$$\mathbf{r}_{\text{com}} = \mathbf{R}_z(\theta) [0, 0, z]^\top \quad (54)$$

Parallel axis

$$\mathbf{I}_{\text{ref}} = \mathbf{I}_{\text{com}} + m (\mathbf{r}_{\text{com}}^\top \mathbf{r}_{\text{com}} \mathbf{I}_3 - \mathbf{r}_{\text{com}} \mathbf{r}_{\text{com}}^\top) \quad (55)$$

0.2.2 Regression

wrench measurement sign is flipped

0.2.3 RMS Error

$$\text{RMS} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{k=1}^{N_{\text{test}}} \left(\mathbf{w}^k - \mathbf{A}^k \phi(\hat{\boldsymbol{\theta}}) - \hat{\mathbf{b}} \right)^2} \in \mathbb{R}^6 \quad (56)$$

$$\text{RMS}_{\%} = \frac{100 \cdot \text{RMS}}{\max(\mathbf{y}) - \min(\mathbf{y})} \quad (57)$$

0.3 Door Body

0.3.1 Sensor Frame in World Frame

Makes vectors from P_2 to P_3 , and P_2 to P_1 . Transformation matrix ${}^S\mathbf{T}_M$ is known.

$$\hat{\mathbf{x}} = \frac{P_3 - P_2}{\|P_3 - P_2\|}, \quad (58)$$

$$\mathbf{v} = \frac{P_1 - P_2}{\|P_1 - P_2\|}, \quad (59)$$

$$\hat{\mathbf{z}} = \frac{\mathbf{v} \times \hat{\mathbf{x}}}{\|\mathbf{v} \times \hat{\mathbf{x}}\|}, \quad (60)$$

$$\hat{\mathbf{y}} = -\hat{\mathbf{x}} \times \hat{\mathbf{z}} \quad (61)$$

$${}^W\mathbf{T}_M^k = \begin{bmatrix} \hat{\mathbf{x}}^k & \hat{\mathbf{y}}^k & \hat{\mathbf{z}}^k & {}^W\mathbf{r}_{W,M}^k \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (62)$$

$${}^W\mathbf{T}_S^k = {}^W\mathbf{T}_M^k {}^S\mathbf{T}_M^{-1} \quad (63)$$

0.3.2 Body Frame in World Frame

$${}^W O_B = \frac{1}{6} \sum_{i=1}^6 {}^W P_{Hi} \quad (64)$$

$$\mathbf{r}_{B,P}^W = {}^W P_{lc} - {}^W O_B \quad (65)$$

$$\hat{\mathbf{z}}_B = \mathbf{v}_{\min}(\text{SVD}(\{{}^W P_{Hi}\} - {}^W O_B)), \quad (66)$$

$$\hat{\mathbf{x}}_B = \frac{\mathbf{r}_{B,P}^W - (\mathbf{r}_{B,P}^W \cdot \hat{\mathbf{z}}_B) \hat{\mathbf{z}}_B}{\|\mathbf{r}_{B,P}^W - (\mathbf{r}_{B,P}^W \cdot \hat{\mathbf{z}}_B) \hat{\mathbf{z}}_B\|}, \quad (67)$$

$$\hat{\mathbf{y}}_B = \hat{\mathbf{z}}_B \times \hat{\mathbf{x}}_B \quad (68)$$

$${}^W \mathbf{T}_B^k = \begin{bmatrix} \hat{\mathbf{x}}_B & \hat{\mathbf{y}}_B & \hat{\mathbf{z}}_B & \mathbf{r}_{W,B}^W \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (69)$$

0.3.3 Handle Wrench Estimation

$${}^S \mathbf{g}^k = {}^W \mathbf{R}_S^{k\top} {}^W \mathbf{g} \quad (70)$$

0.3.4 Static Preload

$$\mathbf{b}_S = \frac{1}{N_{\text{test}}} \sum_{k=1}^{N_{\text{test}}} (\mathbf{w}_S^k - \hat{\mathbf{w}}_S^k) \quad (71)$$

$${}^S \hat{\mathbf{w}}^k = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{S}(\mathbf{r}_h) & \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} m_h {}^S \mathbf{g}^k \\ \mathbf{0}_3 \end{bmatrix} \quad (72)$$

0.3.5 Kinematics and Dynamics

$+\theta$ is negative about world $\mathbf{z}$, so need a sign flip

$$\theta^k = -(\text{unwrap}(\text{atan2}({}^W \mathbf{x}_{B[2]}^k, {}^W \mathbf{x}_{B[1]}^k)) - \theta^1) \quad (73)$$

Savitzky-Golay derivative

$$\dot{\theta} = \mathcal{D}_{\text{SG}}^{(1)}[\theta] \quad (74)$$

$$\ddot{\theta} = \mathcal{D}_{\text{SG}}^{(2)}[\theta] \quad (75)$$

0.3.6 Compute External Torque

Compute wrench at body frame

$${}^B \mathbf{R}_S^k = {}^W \mathbf{R}_B^{k\top} {}^W \mathbf{R}_S^k, \quad (76)$$

$${}^B \mathbf{r}_{B,S}^k = {}^W \mathbf{R}_B^{k\top} ({}^W \mathbf{r}_{W,S}^k - {}^W \mathbf{r}_{W,B}^k) \quad (77)$$

apply 1-DOF constraint

$$\mathbf{w}_B^k = {}^B \mathbf{W}_S^k (\mathbf{w}_S^k - \hat{\mathbf{w}}_S^k - \mathbf{b}_S) \quad (78)$$

$$\tau_{\text{ext}}^k = \mathbf{w}_{B[6]}^k \quad (79)$$

0.3.7 Fourbar Linkage

$$K_1 = \frac{L_4}{L_1}, \quad K_2 = \frac{L_4}{L_2}, \quad K_3 = \frac{L_3^2 - L_4^2 - L_1^2 - L_2^2}{2L_1L_2} \quad (80)$$

$$D = (1 + K_2) \cos \theta - K_1 + K_3, \quad E = -2 \sin \theta, \quad F = K_1 + (K_2 - 1) \cos \theta + K_3 \quad (81)$$

$$\gamma = 2 \arctan \left(\frac{-E - \sqrt{E^2 - 4DF}}{2D} \right) \quad (82)$$

$$\phi = \pi + \theta - \gamma \quad (83)$$

$$\nu = \frac{d\phi}{d\theta} \quad (84)$$

0.3.8 Operating Zones

$$\text{opening} : \dot{\phi} > 0, \theta < \theta_{bc}, \quad \text{backcheck} : \dot{\phi} > 0, \theta \geq \theta_{bc}, \quad (85)$$

$$\text{sweep} : \dot{\phi} < 0, \theta > \theta_l, \quad \text{latch} : \dot{\phi} < 0, \theta \leq \theta_l \quad (86)$$

$$w_{bc}(\theta) = \frac{1}{2} \left(1 + \tanh \frac{\theta - \theta_{bc}}{\varepsilon_{bc}} \right), \quad (87)$$

$$w_l(\theta) = \frac{1}{2} \left(1 + \tanh \frac{\theta_l - \theta}{\varepsilon_l} \right), \quad (88)$$

$$w_s = 1 - w_l \quad (89)$$

0.3.9 Quasi-static Spring Estimation

$$\tau_s^k = \frac{-\tau_{\text{ext}}^k}{\nu^k}, \quad k \in \mathcal{K}_{qs} \quad (90)$$

$$\tau_s(\phi) \approx c_3 \phi^3 + c_2 \phi^2 + c_1 \phi + c_0 \quad (91)$$

0.3.10 Regression Reference

$$y^k = \tau_{\text{ext}}^k + \nu^k \hat{\tau}_s(\phi^k) \quad (92)$$

0.3.11 Regression

$$f = I \ddot{\theta} + b_v \dot{\theta} + b_c \text{sign}(\dot{\theta}) + \nu \tau_b \quad (93)$$

$$\tau_b(\dot{\phi}) = \begin{cases} w_{bc} b_{bc} \dot{\phi}, & \dot{\phi} > 0, \\ (w_s b_s + w_l b_l) \dot{\phi}, & \dot{\phi} < 0, \\ 0, & \dot{\phi} = 0 \end{cases} \quad (94)$$

$$r^k(\theta_I, \boldsymbol{\theta}_H) = y^k - f(\theta_I, \boldsymbol{\theta}_H) \quad (95)$$

$$(\hat{\boldsymbol{\theta}}_I, \hat{\boldsymbol{\theta}}_H) = \arg \min_{\boldsymbol{\theta}_I, \boldsymbol{\theta}_H} \sum_{k=1}^N r^k(\boldsymbol{\theta}_I, \boldsymbol{\theta}_H)^2 \quad (96)$$

0.3.12 RMS Error

$$\text{RMS} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{k=1}^{N_{\text{test}}} r^k(\hat{\boldsymbol{\theta}}_I, \hat{\boldsymbol{\theta}}_H)^2} \quad (97)$$

$$\text{RMS}_{\%} = \frac{100 \cdot \text{RMS}}{\max(\mathbf{y}) - \min(\mathbf{y})} \quad (98)$$

0.4 Wheelbarrow Body

0.4.1 Compute Body Frame in World

$$\hat{\mathbf{z}}_B = \frac{P_2 - P_1}{\|P_2 - P_1\|}, \quad (99)$$

$$\mathbf{y}_{\text{raw}} = \frac{1}{2}[(P_{a2} - P_{a1}) + (P_{b2} - P_{b1})], \quad (100)$$

$$\hat{\mathbf{x}}_B = \frac{\mathbf{y}_{\text{raw}} \times \hat{\mathbf{z}}_B}{\|\mathbf{y}_{\text{raw}} \times \hat{\mathbf{z}}_B\|}, \quad (101)$$

$$\hat{\mathbf{y}}_B = \hat{\mathbf{z}}_B \times \hat{\mathbf{x}}_B \quad (102)$$

$${}^W\mathbf{R}_B^k = [\hat{\mathbf{x}}_B \quad \hat{\mathbf{y}}_B \quad \hat{\mathbf{z}}_B], \quad (103)$$

$${}^W\mathbf{r}_{W,B}^k = \frac{1}{2}(P_1 + P_2) \quad (104)$$

0.4.2 Compute Right Handle Frame in Body Frame

Assumes ${}^W\mathbf{r}_{W,B}^k$, ${}^W\mathbf{R}_B^k$ and ${}^W\mathbf{r}_{W,R}^k$, ${}^W\mathbf{R}_R^k$ already known.

$${}^B\mathbf{R}_R^k = {}^W\mathbf{R}_B^{k\top} {}^W\mathbf{R}_R^k, \quad (105)$$

$${}^B\mathbf{r}_{B,R}^k = {}^W\mathbf{R}_B^{k\top} ({}^W\mathbf{r}_{W,R}^k - {}^W\mathbf{r}_{W,B}^k) \quad (106)$$

0.4.3 Compute Left Handle Frame in Body Frame

Assumes ${}^W\mathbf{r}_{W,B}^k$, ${}^W\mathbf{R}_B^k$ and ${}^W\mathbf{r}_{W,L}^k$, ${}^W\mathbf{R}_L^k$ already known.

$${}^B\mathbf{R}_L^k = {}^W\mathbf{R}_B^{k\top} {}^W\mathbf{R}_L^k, \quad (107)$$

$${}^B\mathbf{r}_{B,L}^k = {}^W\mathbf{R}_B^{k\top} ({}^W\mathbf{r}_{W,L}^k - {}^W\mathbf{r}_{W,B}^k) \quad (108)$$

0.4.4 IMU-Based Dynamics

Compute angular velocity and angular acceleration in B frame

$$\boldsymbol{\omega}_B^k = {}^B\mathbf{R}_R^k \boldsymbol{\omega}_R^k, \quad (109)$$

$$\dot{\boldsymbol{\omega}}_B^k = {}^B\mathbf{R}_R^k \dot{\boldsymbol{\omega}}_R^k \quad (110)$$

Compute linear acceleration at B

$$\mathbf{a}_B^k = {}^B\mathbf{R}_R^k \mathbf{a}_R^k - \dot{\boldsymbol{\omega}}_B^k \times {}^B\mathbf{r}_{B,R}^k - \boldsymbol{\omega}_B^k \times (\boldsymbol{\omega}_B^k \times {}^B\mathbf{r}_{B,R}^k) \quad (111)$$

0.4.5 VICON-Based Dynamics

$$\dot{\mathbf{R}}^k \approx \frac{{}^W\mathbf{R}_B^{k+1} - {}^W\mathbf{R}_B^{k-1}}{2\Delta t} \quad (112)$$

$$\mathbf{S}(\boldsymbol{\omega}^k) = {}^W\mathbf{R}_B^{k\top} \dot{\mathbf{R}}^k \quad (113)$$

$$\boldsymbol{\omega}^k = \begin{bmatrix} \mathbf{S}(\boldsymbol{\omega}^k)_{[3,2]} \\ \mathbf{S}(\boldsymbol{\omega}^k)_{[1,3]} \\ \mathbf{S}(\boldsymbol{\omega}^k)_{[2,1]} \end{bmatrix} \quad (114)$$

Compute linear acceleration at B using VICON data

$$\mathbf{a}_B^{k,\text{VICON}} = {}^W\mathbf{R}_B^{k\top} \left(\mathcal{D}_{\text{SG}}^{(2)} [{}^W\mathbf{r}_{W,B}]^k - {}^W\mathbf{g} \right) \quad (115)$$

Compute angular acceleration and angular velocity in B frame using VICON data

$$\boldsymbol{\omega}_B^{k,\text{VICON}} = \mathcal{D}_{\text{SG}}^{(0)} [\boldsymbol{\omega}_{\text{raw}}]^k, \quad (116)$$

$$\dot{\boldsymbol{\omega}}_B^{k,\text{VICON}} = \mathcal{D}_{\text{SG}}^{(1)} [\boldsymbol{\omega}_{\text{raw}}]^k \quad (117)$$

Compute Scrubbing Rate

$$\dot{\psi}^k = \left({}^W\mathbf{R}_B^k \boldsymbol{\omega}_B^k \right)_{[3]} \quad (118)$$

Compute Contact Frame in World Frame

$${}^W\mathbf{r}_{W,C}^k = \left[{}^W\mathbf{r}_{W,B[1]}^k, {}^W\mathbf{r}_{W,B[2]}^k, 0 \right]^\top \quad (119)$$

$$\hat{\mathbf{z}}_C = [0, 0, 1]^\top, \quad (120)$$

$$\hat{\mathbf{y}}_C = \frac{{}^W\hat{\mathbf{z}}_B^k - ({}^W\hat{\mathbf{z}}_B^k \cdot \hat{\mathbf{z}}_C) \hat{\mathbf{z}}_C}{\| {}^W\hat{\mathbf{z}}_B^k - ({}^W\hat{\mathbf{z}}_B^k \cdot \hat{\mathbf{z}}_C) \hat{\mathbf{z}}_C \|}, \quad (121)$$

$$\hat{\mathbf{x}}_C = \hat{\mathbf{y}}_C \times \hat{\mathbf{z}}_C, \quad (122)$$

$${}^W\mathbf{R}_C^k = [\hat{\mathbf{x}}_C \quad \hat{\mathbf{y}}_C \quad \hat{\mathbf{z}}_C] \quad (123)$$

Compute Body Frame in World Frame

$${}^C\mathbf{R}_B^k = {}^W\mathbf{R}_C^{k\top} {}^W\mathbf{R}_B^k, \quad (124)$$

$${}^C\mathbf{r}_{C,B}^k = {}^W\mathbf{R}_C^{k\top} ({}^W\mathbf{r}_{W,B}^k - {}^W\mathbf{r}_{W,C}^k) \quad (125)$$

0.4.6 Compute Wrench Transmission Matrices

$${}^C\mathbf{W}_B^k = \mathbf{W}({}^C\mathbf{R}_B^k, {}^C\mathbf{r}_{C,B}^k), \quad (126)$$

$${}^C\mathbf{W}_R^k = \mathbf{W}({}^C\mathbf{R}_R^k, {}^C\mathbf{r}_{C,R}^k), \quad (127)$$

$${}^C\mathbf{W}_L^k = \mathbf{W}({}^C\mathbf{R}_L^k, {}^C\mathbf{r}_{C,L}^k) \quad (128)$$

$$\text{pitch}^k = \arcsin({}^W\mathbf{R}_B^k_{[3,2]}), \quad (129)$$

$$\text{roll}^k = \arcsin({}^W\mathbf{R}_B^k_{[3,3]}), \quad (130)$$

$$\text{yaw}^k = \text{atan2}({}^W\mathbf{x}_{C[2]}^k, {}^W\mathbf{x}_{C[1]}^k) \quad (131)$$

0.4.7 Wheel Dynamics

$$\mathbf{v}_C^k = {}^W \mathbf{R}_C^{k\top} \mathcal{D}_{\text{SG}}^{(1)} [{}^W \mathbf{r}_{W,C}]^k \quad (132)$$

$$\Omega_w^k = \frac{\mathbf{v}_{C[1]}^k}{R_w} \quad (133)$$

0.4.8 Regression Reference

$$\mathbf{y}^k = {}^C \mathbf{W}_R^k (\mathbf{w}_R^k - \hat{\mathbf{w}}_R^k - \mathbf{b}_R) + {}^C \mathbf{W}_L^k (\mathbf{w}_L^k - \hat{\mathbf{w}}_L^k - \mathbf{b}_L) \quad (134)$$

0.4.9 Regression

$$\alpha = \frac{1}{2} \log \frac{m}{t_1^2 + t_2^2 + t_3^2 + 1} \quad (135)$$

$$\boldsymbol{\theta}_m = [d_1, d_2, d_3, s_{12}, s_{23}, s_{13}, t_1, t_2, t_3] \in \mathbb{R}^9 \quad (136)$$

$$\boldsymbol{\phi} = \boldsymbol{\phi}(\alpha(\boldsymbol{\theta}_m), \boldsymbol{\theta}_m) \quad (137)$$

$$\boldsymbol{\theta}_f = [\log b_v, \log b_c, \log b_\psi] \in \mathbb{R}^3 \quad (138)$$

$$\mathbf{x} = [\boldsymbol{\theta}_m; \boldsymbol{\theta}_f] \in \mathbb{R}^{12} \quad (139)$$

$$\mathbf{w}_C^k = [0, 0, 0, 0, 0, -b_\psi \dot{\psi}^k]^\top \quad (140)$$

$${}^B \mathbf{w}_A^k = (-b_v \Omega_w^k - b_c \text{sign}(\Omega_w^k)) \mathbf{e}_6 \quad (141)$$

$$\mathbf{f}^k(\mathbf{x}) = {}^C \mathbf{W}_B^k \mathbf{A}^k \boldsymbol{\phi}(\boldsymbol{\theta}_m) - \mathbf{w}_C^k - {}^C \mathbf{W}_B^k {}^B \mathbf{w}_A^k \quad (142)$$

$$\mathbf{C} = [\mathbf{0}_3 \quad \mathbf{I}_3] \quad (143)$$

$$(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\theta}}_f) = \arg \min_{\boldsymbol{\theta}_m, \boldsymbol{\theta}_f} \sum_{k=1}^N \|\mathbf{C} \mathbf{G} (\mathbf{y}^k - \mathbf{f}^k(\boldsymbol{\theta}_m, \boldsymbol{\theta}_f))\|^2 + \lambda^2 \|\boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_{m, \text{CAD}}\|^2 \quad (144)$$

0.4.10 Analytical Jacobian

$$\mathbf{r}(\boldsymbol{\theta}_m, \boldsymbol{\theta}_f) = \begin{bmatrix} \mathbf{C} \mathbf{G} (\mathbf{y}^k - \mathbf{f}^k(\boldsymbol{\theta}_m, \boldsymbol{\theta}_f)) \\ \lambda (\boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_{m, \text{CAD}}) \end{bmatrix} \quad (145)$$

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \mathbf{r}}{\partial \boldsymbol{\theta}_m} & \frac{\partial \mathbf{r}}{\partial \boldsymbol{\theta}_f} \end{bmatrix} = \left[\begin{array}{c|c} -\mathbf{C} \mathbf{G} {}^C \mathbf{W}_B^k \mathbf{A}^k \frac{\partial \boldsymbol{\phi}}{\partial \boldsymbol{\theta}_m} & -\mathbf{C} \mathbf{G} \frac{\partial \mathbf{f}^k}{\partial \boldsymbol{\theta}_f} \\ \hline \lambda \mathbf{I} & \mathbf{0} \end{array} \right] \quad (146)$$

$$\frac{\partial \mathbf{f}^k}{\partial \boldsymbol{\theta}_f} = [\Omega_w^k {}^C \mathbf{W}_B^k \mathbf{e}_6 b_v \quad \text{sign}(\Omega_w^k) {}^C \mathbf{W}_B^k \mathbf{e}_6 b_c \quad \dot{\psi}^k \mathbf{e}_6 b_\psi] \quad (147)$$

$$\boldsymbol{\phi} = \frac{m_{\text{known}}}{D} \mathbf{n}(\boldsymbol{\theta}_m) \quad (148)$$

Applying the product rule:

$$\frac{\partial \boldsymbol{\phi}}{\partial \boldsymbol{\theta}_m} = \frac{m_{\text{known}}}{D} \frac{\partial \mathbf{n}}{\partial \boldsymbol{\theta}_m} + \mathbf{n} \left(\frac{\partial}{\partial \boldsymbol{\theta}_m} \frac{m_{\text{known}}}{D} \right)^\top \quad (149)$$

Since $D = t_1^2 + t_2^2 + t_3^2 + 1$ depends only on t_1, t_2, t_3 , the second term is zero for the first six parameters:

$$\frac{\partial}{\partial \boldsymbol{\theta}_m} \frac{m_{\text{known}}}{D} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\frac{2m_{\text{known}} t_1}{D^2} \\ -\frac{2m_{\text{known}} t_2}{D^2} \\ -\frac{2m_{\text{known}} t_3}{D^2} \end{bmatrix} \quad (150)$$

$$\frac{\partial \phi}{\partial \boldsymbol{\theta}_m} = \frac{m_{\text{known}}}{D} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 2t_1 & 2t_2 & 2t_3 \\ e^{d_1} t_1 & 0 & 0 & 0 & 0 & 0 & e^{d_1} & 0 & 0 \\ 0 & e^{d_2} t_2 & 0 & t_1 & 0 & 0 & s_{12} & e^{d_2} & 0 \\ 0 & 0 & e^{d_3} t_3 & 0 & t_2 & t_1 & s_{13} & s_{23} & e^{d_3} \\ 0 & 2e^{2d_2} & 2e^{2d_3} & 2s_{12} & 2s_{23} & 2s_{13} & 0 & 0 & 0 \\ -e^{d_1} s_{12} & 0 & 0 & -e^{d_1} & 0 & 0 & 0 & 0 & 0 \\ -e^{d_1} s_{13} & 0 & 0 & 0 & 0 & -e^{d_1} & 0 & 0 & 0 \\ 2e^{2d_1} & 0 & 2e^{2d_3} & 0 & 2s_{23} & 2s_{13} & 0 & 0 & 0 \\ 0 & -e^{d_2} s_{23} & 0 & -s_{13} & -e^{d_2} & -s_{12} & 0 & 0 & 0 \\ 2e^{2d_1} & 2e^{2d_2} & 0 & 2s_{12} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} - \frac{2}{D} \phi \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ t_1 \\ t_2 \\ t_3 \end{bmatrix}^\top \quad (151)$$

0.4.11 Condition Number

$$\kappa_{\text{eff}} = \sqrt{\frac{s_1^2 + \lambda^2}{s_9^2 + \lambda^2}} \quad (152)$$

0.4.12 RMS Error

$$\text{RMS} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{k=1}^{N_{\text{test}}} \left(\mathbf{C} \left(\mathbf{y}^k - \mathbf{f}^k(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\theta}}_f) - \mathbf{b} \right) \right)^2} \in \mathbb{R}^3 \quad (153)$$

$$\text{RMS}_{\%} = \frac{100 \cdot \text{RMS}}{\max(\mathbf{y}) - \min(\mathbf{y})} \quad (154)$$